Practical No. 05

ON

SSCA4030 - Image Processing

TITLE: **To implement the Fourier Transform of an image using Python and visualize its frequency components.**

BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY (HONOURS.)

SUBMITTED BY:

Name: Shah Rishabh Alpeshbhai

Enrolment: 23SS02IT181

Class: BSCIT-H – 7A - Batch 2023-27

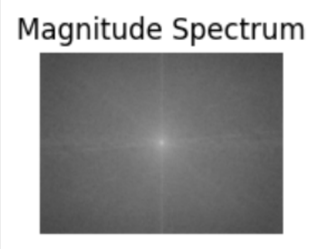
Aim: The Fourier Transform (FT) converts an image from the spatial domain to the frequency domain. Low-frequency components represent smooth regions and overall brightness. High-frequency components represent edges, fine details, and noise. The Discrete Fourier Transform (DFT) is computed using the Fast Fourier Transform (FFT) algorithm.

Input Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread('/content/image.webp', 0)
f = np.fft.fft2(img)
fshift = np.fft.fftshift(f)
magnitude_spectrum = 20 * np.log(np.abs(fshift) + 1)
f_ishift = np.fft.ifftshift(fshift)
img_back = np.fft.ifft2(f_ishift)
img_back = np.abs(img_back)
plt.figure(figsize=(12,4))
plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')
```


```
plt.subplot(1,3,2)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(735.5), np.float64(551.5), np.float64(-0.5))
```



plt.subplot(1,3,3)

plt.imshow(img_back, cmap='gray')

plt.title("Reconstructed Image")

plt.axis('off')

```
(np.float64(-0.5), np.float64(735.5), np.float64(551.5), np.float64(-0.5))
```

plt.tight_layout()

plt.show()

```
<Figure size 640x480 with 0 Axes>
```